

Khawaja Muhammad Ali

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Education

National University of Science and Technology (NUST)	B.S. Computer Science • GPA: 3.52/4.0
Computer Science	Islamabad, Pakistan • Expected May 2028

Experience

TUKL Laboratory	Islamabad, Pakistan
AI Research Intern	Jun 2025 – Sep 2025

- Built a U-Net++ (**EfficientNet-B4**) based segmentation model for plant disease detection, achieving robust localization of infected regions via optimized preprocessing and augmentation on the NWRD dataset, resulting in a 14% improvement on the accuracy from the previous best.

<https://huggingface.co/KM-Alee/nwr-d-tukl>

SiBot Technologies	Islamabad
AI Engineer Intern	Feb 2026 - Present

- Built a local-first **AI chatbot for NUST admissions** and information queries using **RAG** with Ollama-based **LLMs**, while researching and applying **secure on-device deployment pipelines** and privacy-preserving architectures to improve response accuracy and relevance.

Projects

Tytion

- Built a real-time Typst editing platform using **Next.js**, **React**, **Node.js**, **PostgreSQL**, and **Prisma**, integrating **JWT**-based authentication with live preview and extensible block-based components.
- Delivered advanced UX with drag-and-drop layout building, AI-driven reusable component generation, template synthesis, and automated code refinement.

Llama Studio

- Built a high-performance, local-first AI studio for **llama.cpp** using Rust (Axum), enabling real-time chat, GGUF model management, and advanced inference controls.
- Engineered a privacy-first LLM system with subprocess orchestration, live performance metrics (tokens/sec, VRAM), and CI-tested architecture for stability and scalability.

Masmide

- Engineered a **high-performance TUI IDE for MASM assembly** using Rust and Ratatui, optimizing developer experience in a terminal environment
- Built **deep Linux integrations** including process orchestration, toolchain execution, and low-level system interactions for seamless assembly development

Secure Attendance System

- Architected full-stack attendance system with **4-layer verification** process using **Flutter** mobile app, **React** web dashboard, and **FastAPI** microservices backend with **PostgreSQL** database
- Implemented **Google ML Kit face recognition**, geofencing validation, and deployed scalable infrastructure with **Docker** and **Redis** caching for real-time verification

Technical Skills

Tools & Frameworks

Docker, AWS, Firebase, Git, Linux, Nginx, NumPy, Pandas, React, Node, Flutter, Redis

Languages

Python, TypeScript, Go, Dart, C++, Java, Rust